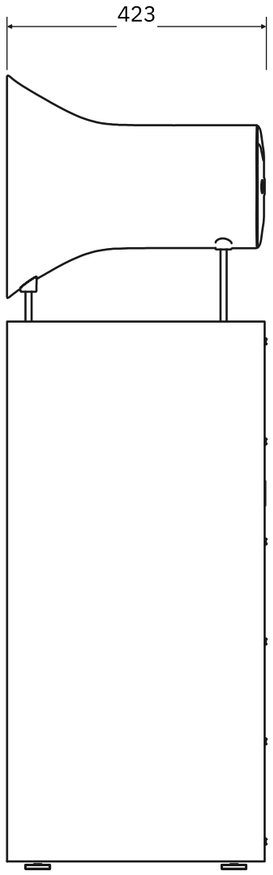
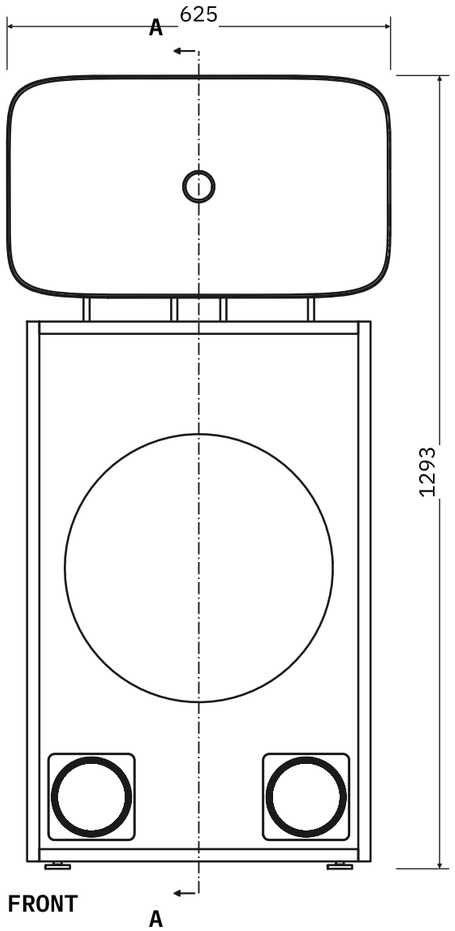
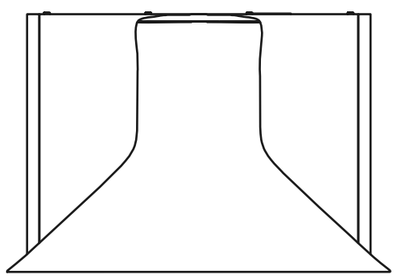
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

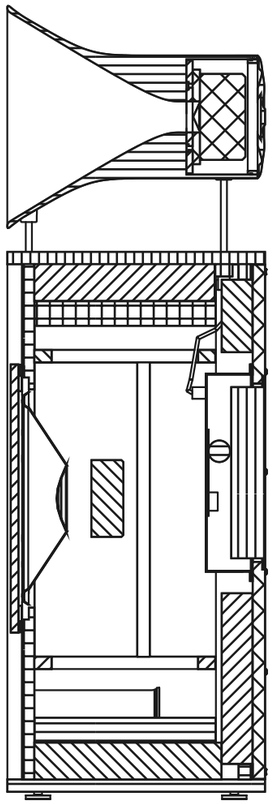
General arrangement

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Section A-A (sheet 1): the product cut at x 280.0 and seen from the right; cut faces hatched, a different angle for each part.



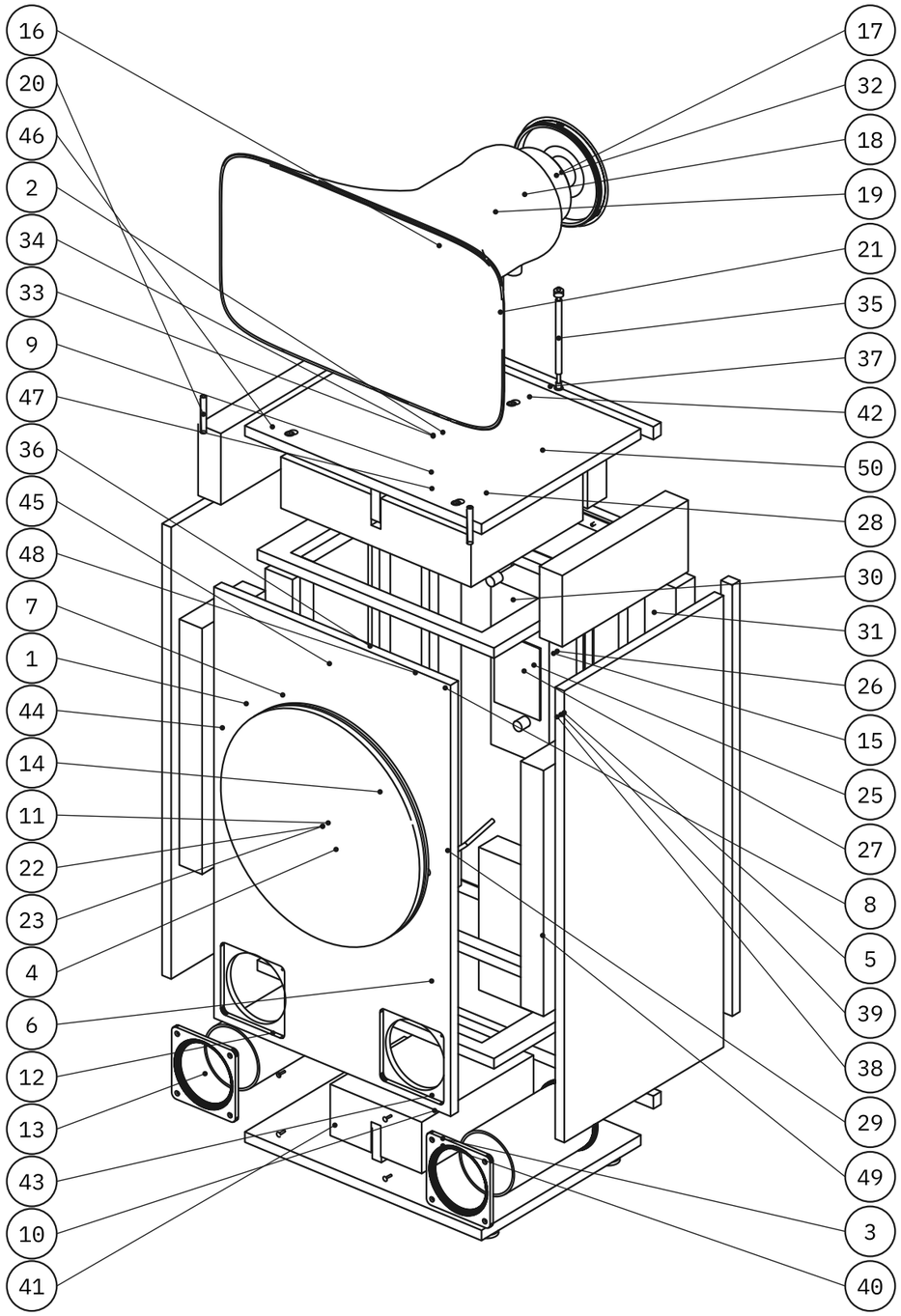
Section A-A

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



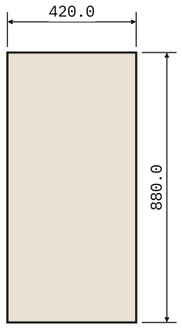
Exploded view

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

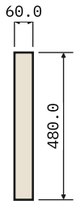
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:20. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



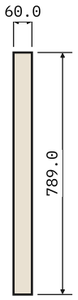
left, right

20 mm cast-acrylic, 4 for the set



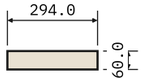
rib-left-1, rib-right-1

20 mm cast-acrylic, 4 for the set



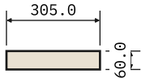
rib-back-1, rib-back-2

20 mm cast-acrylic, 4 for the set



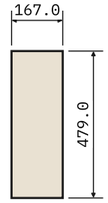
rib-top

20 mm cast-acrylic, 2 for the set



rib-bottom

20 mm cast-acrylic, 2 for the set



lining-left-2a, lining-right-2a

50 mm polyester-batting, 4 for the set

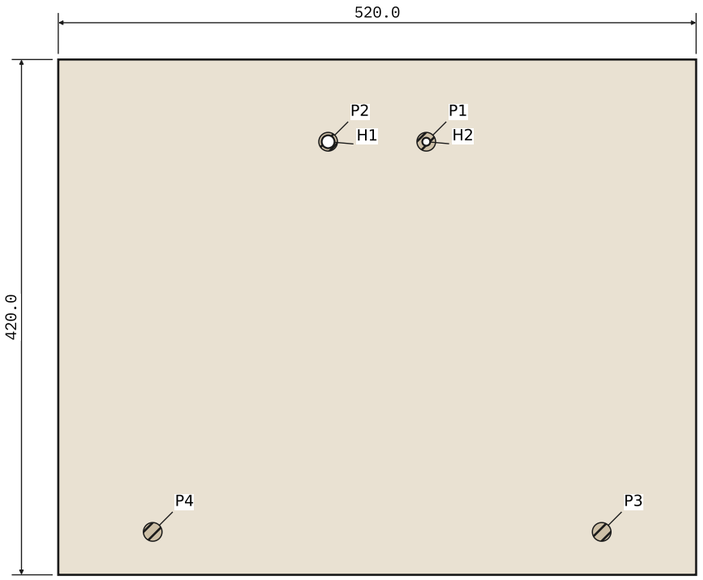
Sheet parts: plain panels

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/top.dxf. Marks listed at the right.



top: 20 mm cast-acrylic, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø15.2	300.0	353.0	15 from face
P2	Ø15.2	220.0	353.0	15 from face
P3	Ø15.2	443.0	35.0	15 from face
P4	Ø15.2	77.0	35.0	15 from face
H1	Ø10.5	220.0	353.0	through
H2	Ø6.5	300.0	353.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

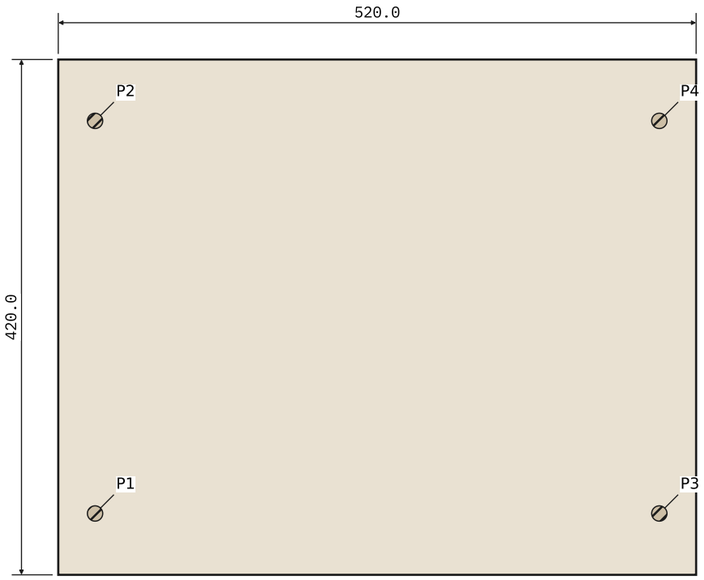
Sheet part: top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/bottom.dxf. Marks listed at the right.



bottom: 20 mm cast-acrylic, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø12.5	30.0	50.0	15 from face
P2	Ø12.5	30.0	370.0	15 from face
P3	Ø12.5	490.0	50.0	15 from face
P4	Ø12.5	490.0	370.0	15 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

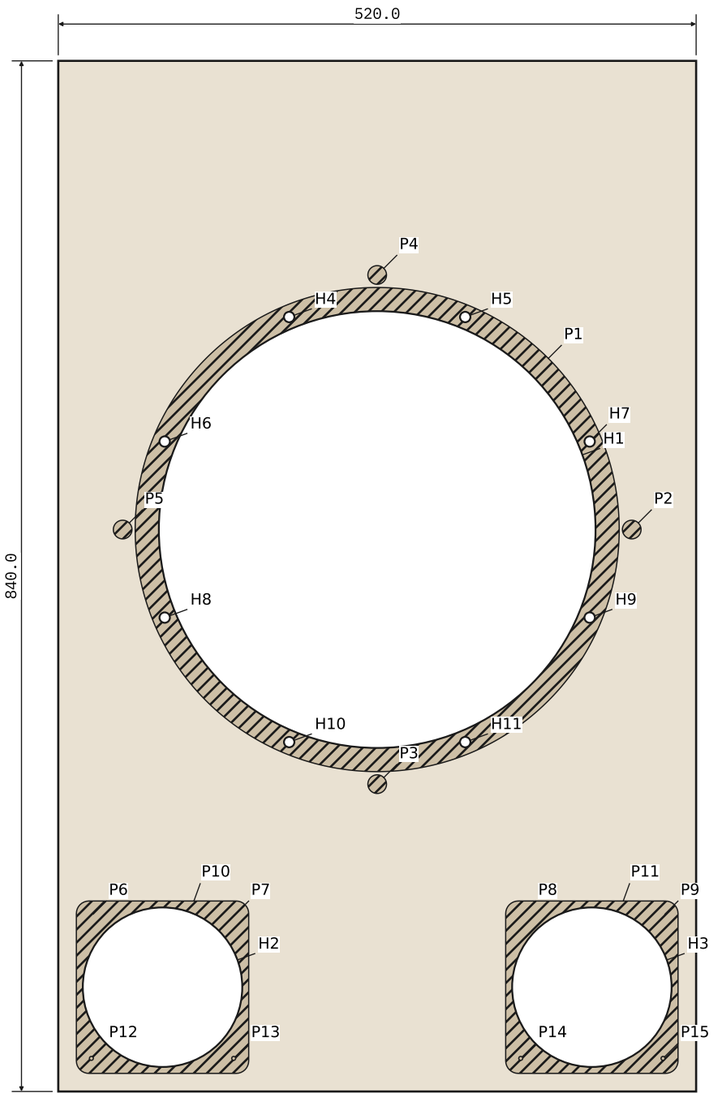
Sheet part: bottom

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/front-panel.dxf. Marks listed at the right.



front-panel: 20 mm cast-acrylic, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø394.6	260.0	458.0	11 from face
P2	Ø15.2	467.5	458.0	3.2 from face
P3	Ø15.2	260.0	250.5	3.2 from face
P4	Ø15.2	260.0	665.5	3.2 from face
P5	Ø15.2	52.5	458.0	3.2 from face
P6	profile (DXF)	27.0	143.0	17 from face
P7	profile (DXF)	143.0	143.0	17 from face
P8	profile (DXF)	377.0	143.0	17 from face
P9	profile (DXF)	493.0	143.0	17 from face
P10	profile (DXF)	85.0	85.0	9 from face
P11	profile (DXF)	435.0	85.0	9 from face
P12	profile (DXF)	27.0	27.0	17 from face
P13	profile (DXF)	143.0	27.0	17 from face
P14	profile (DXF)	377.0	27.0	17 from face
P15	profile (DXF)	493.0	27.0	17 from face
H1	Ø356.0	260.0	458.0	through
H2	Ø130.0	85.0	85.0	through
H3	Ø130.0	435.0	85.0	through
H4	Ø8.5	188.2	631.2	through
H5	Ø8.5	331.8	631.2	through
H6	Ø8.5	86.8	529.8	through
H7	Ø8.5	433.2	529.8	through
H8	Ø8.5	86.8	386.2	through
H9	Ø8.5	433.2	386.2	through
H10	Ø8.5	188.2	284.8	through
H11	Ø8.5	331.8	284.8	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

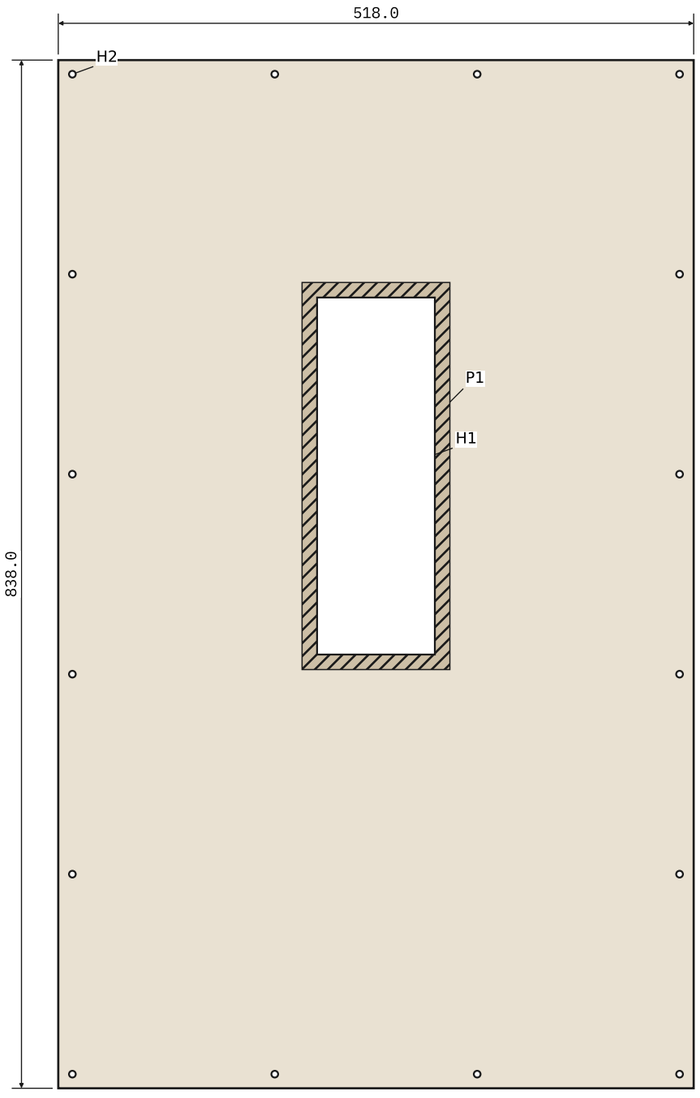
Sheet part: front-panel

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/back.dxf. Marks listed at the right.



back: 20 mm cast-acrylic, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	120.5 x 315.5	259.0	499.0	4.5 from face
H1	96.0 x 291.0	259.0	499.0	through
H2	Ø5.5	16 off	DXF	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: back

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

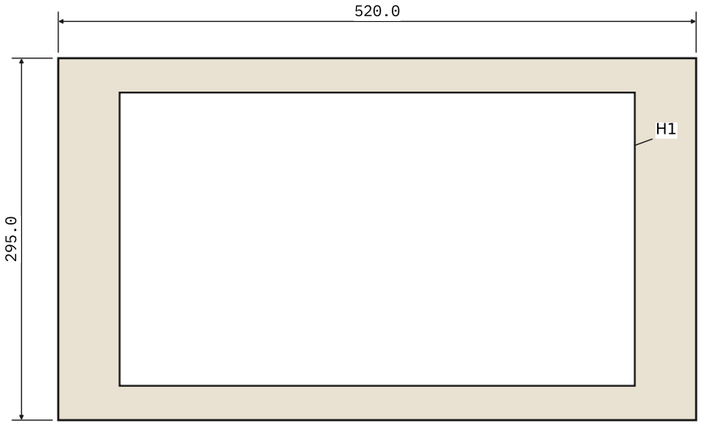
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	420.0 x 239.0	260.0	147.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



brace: 20 mm cast-acrylic, 2 for the set, seen from its outer face

Sheet part: brace

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

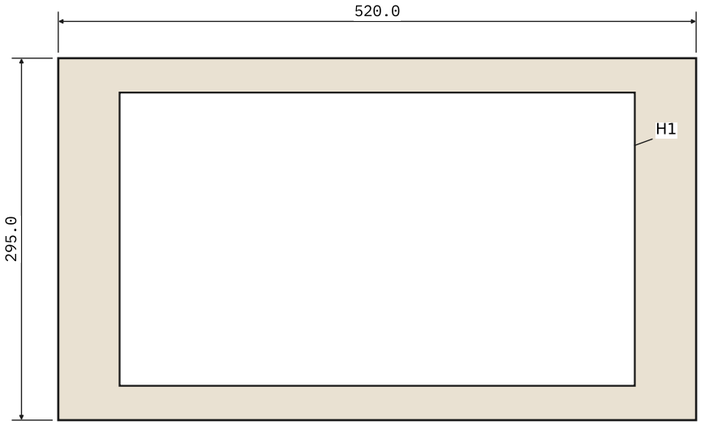
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace-2.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	420.0 x 239.0	260.0	147.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



brace-2: 20 mm cast-acrylic, 2 for the set, seen from its outer face

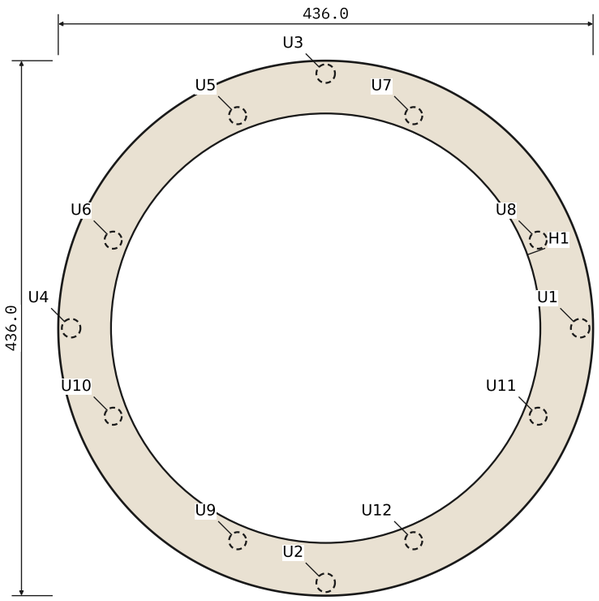
Sheet part: brace-2

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/woofer-ring.dxf. Marks listed at the right.



woofer-ring: 6 mm cast-acrylic, 2 for the set, seen from its outer face

inner-face pockets: dxf/woofer-ring-underside.dxf (seen from that face)

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	425.5	218.0	3.2 from inside
U2	Ø15.2	218.0	10.5	3.2 from inside
U3	Ø15.2	218.0	425.5	3.2 from inside
U4	Ø15.2	10.5	218.0	3.2 from inside
U5	Ø14.0	146.3	391.2	4 from inside
U6	Ø14.0	44.8	289.8	4 from inside
U7	Ø14.0	289.8	391.2	4 from inside
U8	Ø14.0	391.2	289.8	4 from inside
U9	Ø14.0	146.3	44.8	4 from inside
U10	Ø14.0	44.8	146.2	4 from inside
U11	Ø14.0	391.2	146.2	4 from inside
U12	Ø14.0	289.8	44.8	4 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

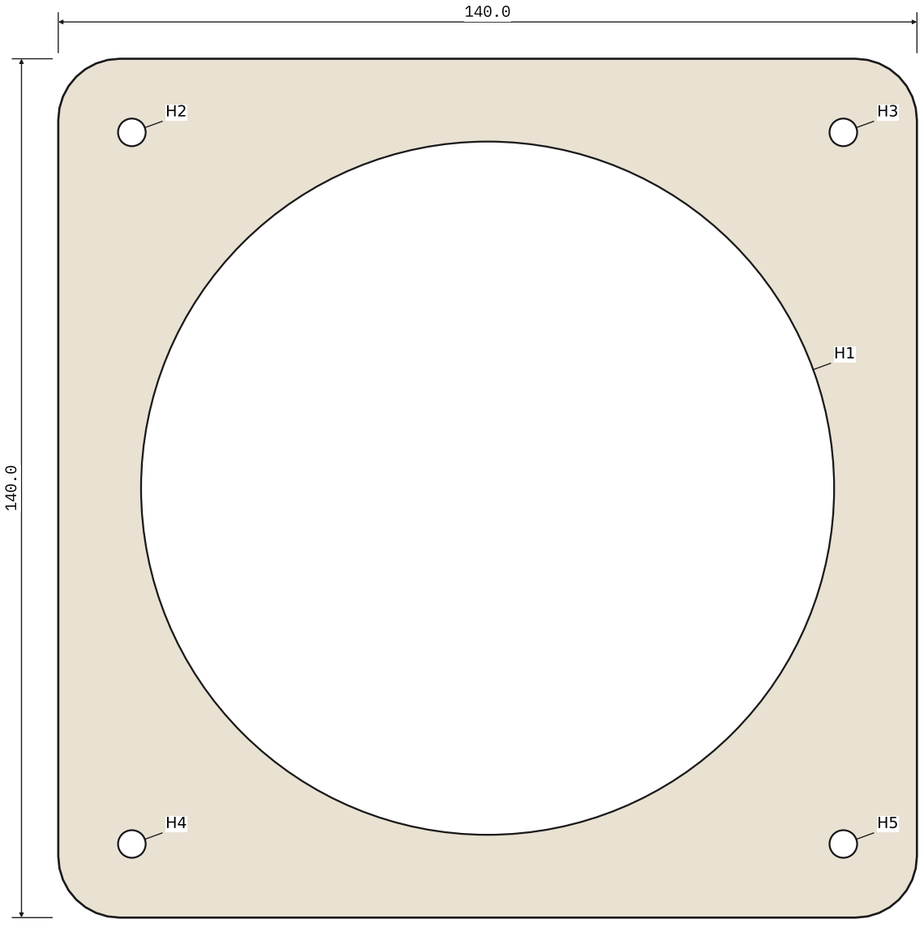
Sheet part: woofer-ring

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/port-frame-left.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	Ø113.0	70.0	70.0	through
H2	Ø4.5	12.0	128.0	through
H3	Ø4.5	128.0	128.0	through
H4	Ø4.5	12.0	12.0	through
H5	Ø4.5	128.0	12.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

port-frame-left: 8 mm cast-acrylic, 2 for the set, seen from its outer face

not in the cut file: 4 holes countersunk 90 deg from the outer face
and: the 113 hole at 70, 70 rounded over from the outer face (24 faces, 2 to 88 deg off the sheet)

Sheet part: port-frame-left

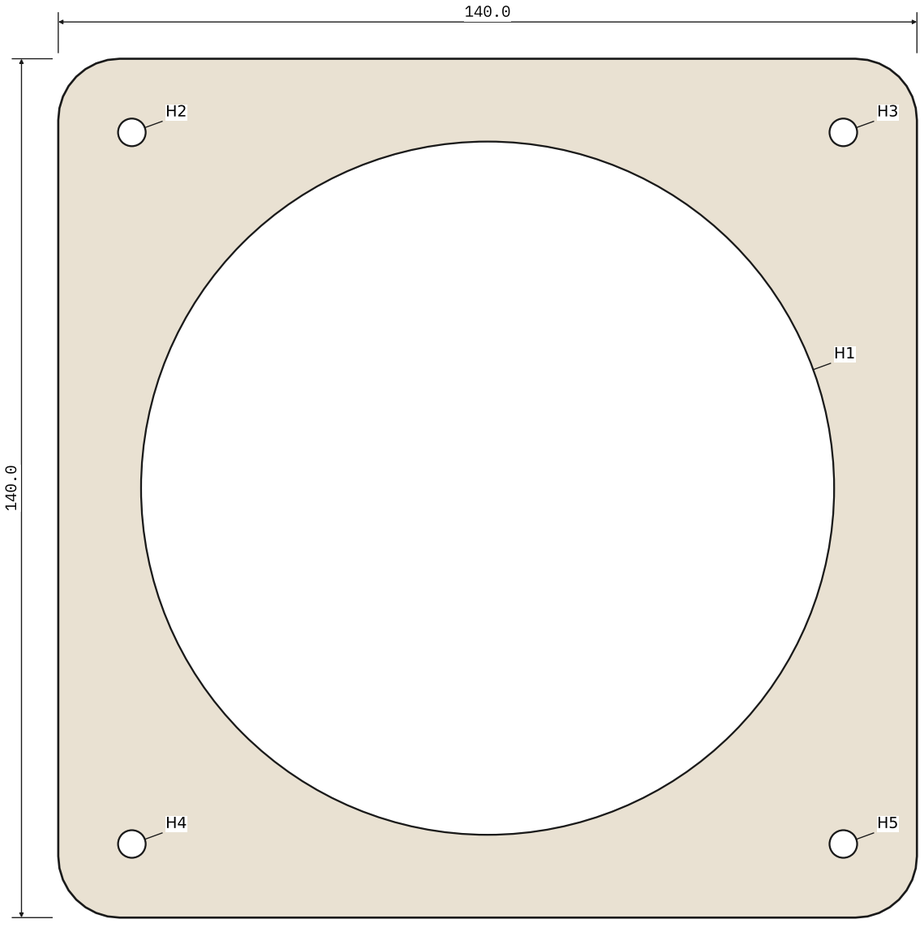
barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

BHC-012 rev A

Scale 1:1. As the router sees it from the outer face; DXF: dxf/port-frame-right.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	Ø113.0	70.0	70.0	through
H2	Ø4.5	12.0	128.0	through
H3	Ø4.5	128.0	128.0	through
H4	Ø4.5	12.0	12.0	through
H5	Ø4.5	128.0	12.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

port-frame-right: 8 mm cast-acrylic, 2 for the set, seen from its outer face

not in the cut file: 4 holes countersunk 90 deg from the outer face
and: the 113 hole at 70, 70 rounded over from the outer face (24 faces, 2 to 88 deg off the sheet)

Sheet part: port-frame-right

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

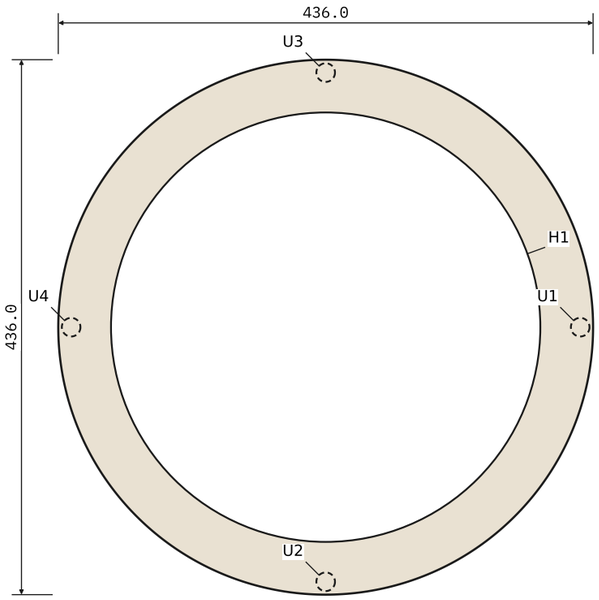
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

BHC-013 rev A

Scale 1:5. As the router sees it from the outer face; DXF: dxf/disc-frame.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	425.5	218.0	3.2 from inside
U2	Ø15.2	218.0	10.5	3.2 from inside
U3	Ø15.2	218.0	425.5	3.2 from inside
U4	Ø15.2	10.5	218.0	3.2 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-frame: 12 mm cast-acrylic, 2 for the set, seen from its outer face

inner-face pockets: dxf/disc-frame-underside.dxf (seen from that face)

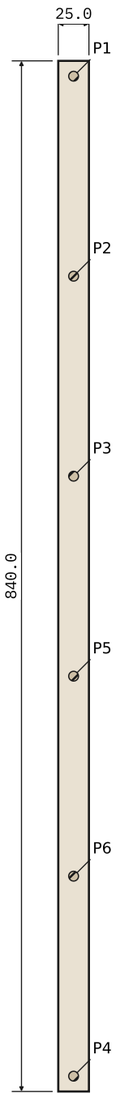
Sheet part: disc-frame

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/land-left.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	Ø8.0	12.5	827.5	12 from face
P2	Ø8.0	12.5	664.5	12 from face
P3	Ø8.0	12.5	501.5	12 from face
P4	Ø8.0	12.5	12.5	12 from face
P5	Ø8.0	12.5	338.5	12 from face
P6	Ø8.0	12.5	175.5	12 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

land-left: 20 mm cast-acrylic, 2 for the set, seen from its outer face

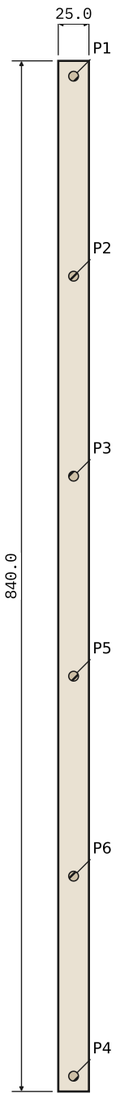
Sheet part: land-left

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/land-right.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	Ø8.0	12.5	827.5	12 from face
P2	Ø8.0	12.5	664.5	12 from face
P3	Ø8.0	12.5	501.5	12 from face
P4	Ø8.0	12.5	12.5	12 from face
P5	Ø8.0	12.5	338.5	12 from face
P6	Ø8.0	12.5	175.5	12 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

land-right: 20 mm cast-acrylic, 2 for the set, seen from its outer face

Sheet part: land-right

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

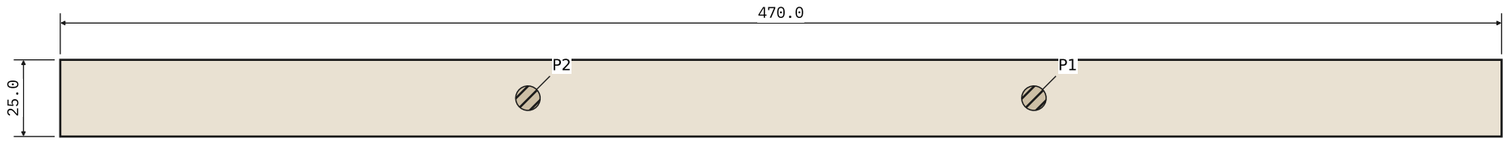
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/land-top.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø8.0	317.5	12.5	12 from face
P2	Ø8.0	152.5	12.5	12 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



land-top: 20 mm cast-acrylic, 2 for the set, seen from its outer face

Sheet part: land-top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

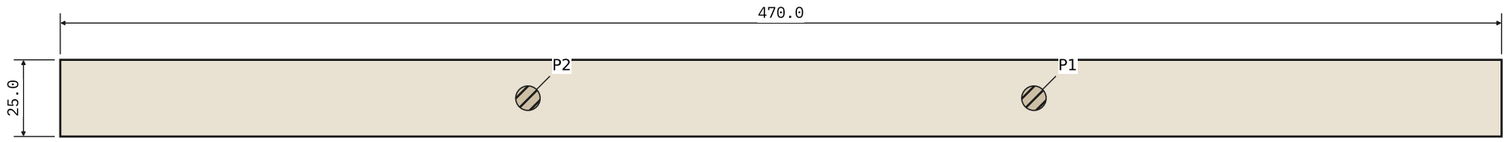
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/land-bottom.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø8.0	317.5	12.5	12 from face
P2	Ø8.0	152.5	12.5	12 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



land-bottom: 20 mm cast-acrylic, 2 for the set, seen from its outer face

Sheet part: land-bottom

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

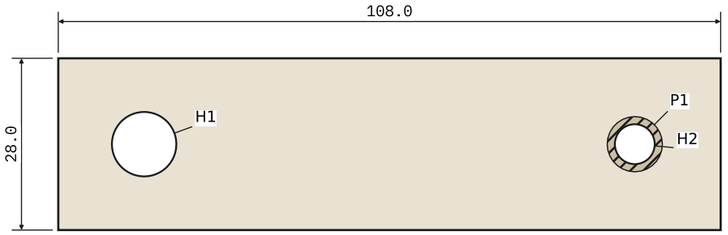
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/top-pad.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø9.0	94.0	14.0	10 from face
H1	Ø10.5	14.0	14.0	through
H2	Ø6.5	94.0	14.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



top-pad: 20 mm cast-acrylic, 2 for the set, seen from its outer face

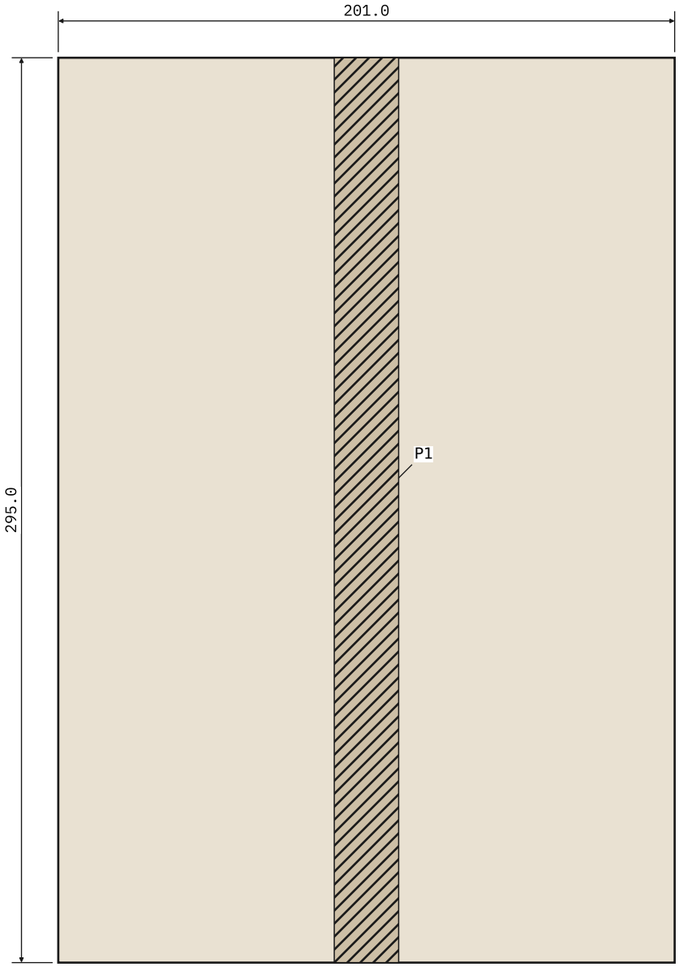
Sheet part: top-pad

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/lining-bottom-1.dxf. Marks listed at the right.



lining-bottom-1: 100 mm polyester-batting, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	21.0 x 295.0	100.5	147.5	60 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

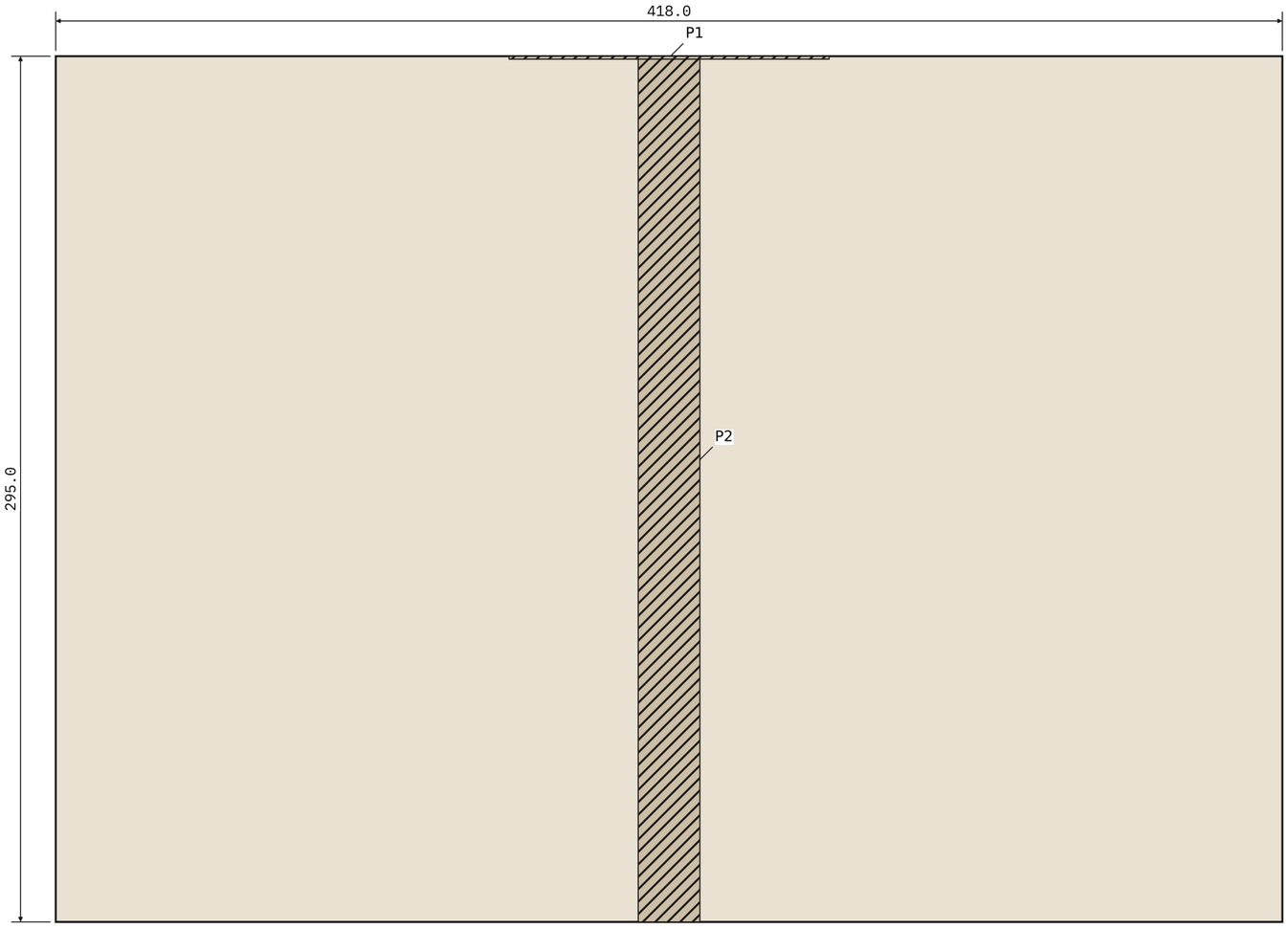
Sheet part: lining-bottom-1

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/lining-top-3.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	109.0 x 1.0	209.0	294.5	20 from face
P2	21.0 x 294.0	209.0	147.0	60 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

lining-top-3: 100 mm polyester-batting, 2 for the set, seen from its outer face

Sheet part: lining-top-3

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

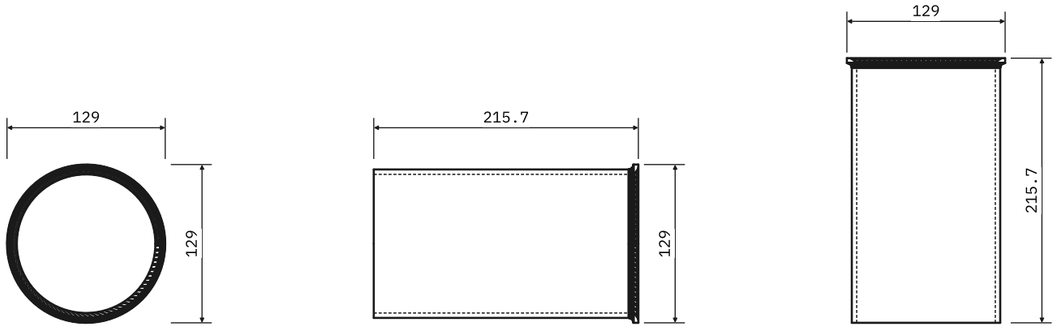
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

12. port-left, port-right: formed, cast-acrylic; 320 cm3 each; 4 for the set; 1:5

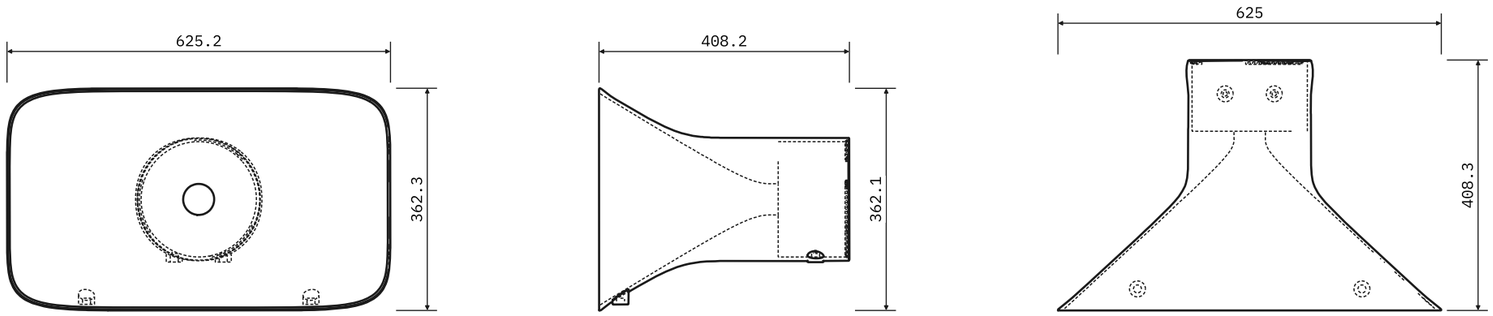
cast acrylic tube 121 x 4 (113 bore, polished), 216 long behind its frame, its inner end flared to R8 over a heated mandrel and annealed; cemented square into its frame's back face; the 2 together tune the box to 38 Hz; its length found on the first sample (three made 10 apart, the one whose impedance minimum falls at fb kept) and the rest made to it: its rounded ends cannot be trimmed



FRONT FROM THE RIGHT FROM ABOVE

16. waveguide: cast, cast-resin; 5218 cm3 each; 2 for the set; 1:10

mouth 610 x 346, 278 deep, 2 in throat; a 10 rim rounded right round, 625 x 362 outside; a shell 6 thick following its flare; the driver in a pod 200 across behind, closed by a cap; four pads under it take the posts' brass inserts; cast in tinted clear resin in a two-part silicone mould from a CNC-cut plug, its faces polished satin (a slow deep-pour resin: the neck round the throat is solid, 5.2 L in all), or pigmented the same way for the opaque and marbled finishes



FRONT FROM THE RIGHT FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

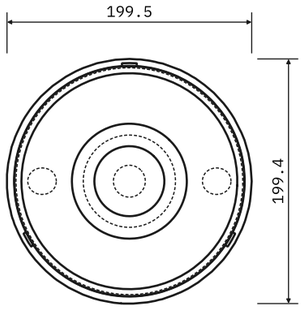
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

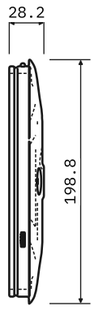
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

17. waveguide-cap: cast, cast-resin; 204 cm3 each; 2 for the set; 1:5

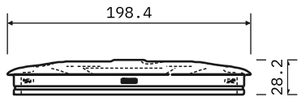
closes the pod behind the driver on a twist-lock: three lugs cast on its spigot go into L-slots in the pod's bore and an eighth of a turn (45 degrees, clockwise from behind) locks them, a small bump in each slot clicking them home; an O-ring on the spigot, beyond the lugs, seals it and stops it rattling; two shallow finger dimples to turn it by; no screw shows; the HF pair leaves through a post below; unplugged here, the horn lifts off its sockets



FRONT



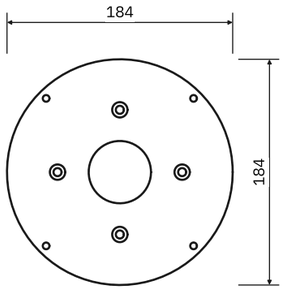
FROM THE RIGHT



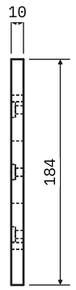
FROM ABOVE

19. hf-adaptor: lathe, aluminium; 242 cm3 each; 2 for the set; 1:5

184 across, 10 thick, bored 50.8 (the throat); 4 countersunk M6 on 101.6 take the driver from the front; 4 x 5.5 on 170 (at 45 degrees to them) take M5 socket caps into the brass inserts in the horn's boss; black anodised



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

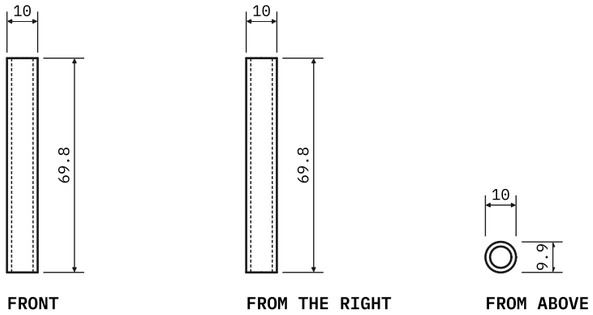
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

20. post-1, post-2: lathe, stainless; 3 cm3 each; 4 for the set; 1:2

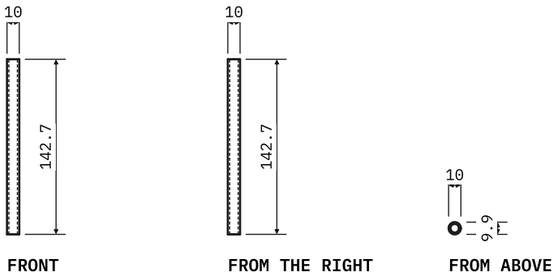
70 long (49 showing): stainless tube 10 x 7, threaded M10 x 1 8 at the top into the brass insert in its pad; its foot plain, standing 13 deep in a polyurethane sleeve in a brass socket in the top; its bore deburred



FRONT FROM THE RIGHT FROM ABOVE

21. post-3, post-4: lathe, stainless; 6 cm3 each; 4 for the set; 1:5

143 long (118 showing): stainless tube 10 x 7, threaded M10 x 1 12 at the top into the brass insert in its pad; its foot plain, standing 13 deep in a polyurethane sleeve in a brass socket in the top; its bore deburred; the HF lead runs down inside it



FRONT FROM THE RIGHT FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

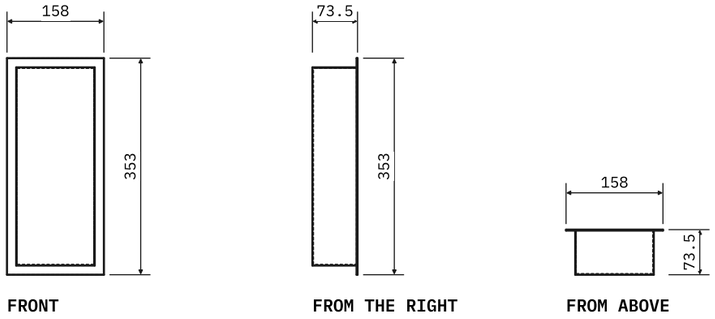
mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

25. amp-box: sheet metal, aluminium; 180 cm3 each; 2 for the set; 1:10

125 x 320 inside, 75 deep from the back: 1.5 aluminium folded, its corners welded, black anodised; a 15 flange round its open side, on a 3 EPDM gasket on the back's inner face, ten screws through it, so the box stays airtight and the mains electronics are enclosed in metal; two M12 cable glands in its front; the ports' length found with it fitted (it takes volume)



Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	left, right	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	420 x 880	175.56
2	top	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	520 x 420	103.74
3	bottom	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	520 x 420	103.74
4	front-panel	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	520 x 840	207.48
5	back	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	518 x 838	206.19
6	brace, brace-2	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	520 x 295	72.87
7	rib-left-1, rib-right-1	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	60 x 480	13.68
8	rib-back-1, rib-back-2	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	60 x 789	22.49
9	rib-top	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	294 x 60	8.38
10	rib-bottom	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	305 x 60	8.69
11	woofer-ring	cut	2	6 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	436 x 436	27.09
12	port-left, port-right	formed	4	Cast acrylic tube, tinted through	320 cm3, 139 x 226 x 139	81.57
13	port-frame-left, port-frame-right	cut	4	8 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	140 x 140	3.72
14	woofer	buy	2	FaitalPRO 15PR400-8 15 in woofer (treated paper cone, neodymium, 8 ohm)	393 frame, 356 cut-out, 165 deep	194.00
15	amp-plate	buy	2	Hypex FusionAmp FA122 DSP plate amplifier, 2 x 125 W into 4 ohm	315 x 120 x 55 plate, 291 x 96 recess	422.40
16	waveguide	cast	2	Casting resin (polyurethane or epoxy), tinted clear, translucent or pigmented	5218 cm3, 635 x 418 x 372	684.59
17	waveguide-cap	cast	2	Casting resin (polyurethane or epoxy), tinted clear, translucent or pigmented	204 cm3, 209 x 38 x 209	138.97
18	hf	buy	2	B&C Speakers DE1085TN 2 in compression driver, titanium diaphragm, 100 mm voice coil, neodymium, 16 ohm (round 5: the owner's bigger throat)	154 across, 86 deep, 4 x M6 on 102	512.00
19	hf-adaptor	lathe	2	Aluminium 6061	194 x 20 x 194	43.82
20	post-1, post-2	lathe	4	Stainless steel 304	20 x 20 x 80	15.96
21	post-3, post-4	lathe	4	Stainless steel 304	20 x 20 x 153	16.83
22	disc-frame	cut	2	12 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	436 x 436	54.18
23	cloth-disc	buy	2	Acoustically transparent speaker cloth (a knit sold for grilles), about 0.6 x 0.95 m a grille, in the finish's colour		15.00
24	cloth-disc-magnets	buy	2	The ring's and the disc's magnets: eight 15 x 3 neodymium discs (N42) and four 15 x 3 mild-steel discs, a set: four magnets in the panel and the steel discs in the ring's back hold the ring, four in the disc's frame hold the disc to the same		10.00
25	amp-box	sheet metal	2	Aluminium 6061	305 x 500 x 2	85.72
26	amp-box-gasket	buy	2	EPDM sponge gasket, 3 mm, cut as a frame 15 wide for the amplifier compartment's flange		6.00
27	hf-pad	buy	2	The HF channel's L-pad: two aluminium-housed wirewound resistors, 25 W (Vishay Dale RH-25 or Arcol HS25 kind), 10 ohm in series and 8.2 ohm across the driver (about -9 dB at 16 ohm; trimmed after the first measurement), screwed with thermal		24.00
28	lead-hf	buy	2	HF lead: round silicone-sheathed cable 2 x 0.5 mm2, about 5 across, 1.3 m, ferrules at the pod's end and a two-way lever connector (Wago 221-412 or like) on the driver's own leads, a polarised 2-way connector at the compartment's end		6.00

Parts list (1 of 3)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
29	lead-woofer	buy	2	Round-sheathed speaker cable 2 x 1.5 mm2, 1 m		2.00
30	cable-glands	buy	2	Three M12 nylon cable glands for 3 to 6.5 mm cable, black: two in the amplifier compartment's front, one under the top at the cable post, in an M12 insert for plastics (in acrylic, a thread tapped in the top)		4.00
31	rating-plate	buy	2	Rating plate 70 x 40 x 1, black anodised aluminium, laser-marked: the model, the serial number, the ratings, the marks (CE, UKCA, FCC, the WEEE bin) and the maker's address; two M3 screws into inserts		10.00
32	cap-o-ring	buy	2	O-ring for the horn's cap, 182 x 2 mm, EPDM 70 Shore A (round 16: the twist-lock), and a tube of silicone grease		4.00
33	post-sockets	buy	2	The posts' brass fittings: four M10 x 1 threaded inserts bonded with epoxy into the waveguide's pads after casting, in holes drilled through a jig taken off the top's sockets (13 across; the front pair 10 long, the rear 14, the tie post's		48.00
34	post-sleeves	buy	2	Polyurethane sleeves 12 x 10 x 13, Shore A 90, a set of four: pressed into the posts' sockets, they centre the posts and stop them buzzing		8.00
35	tie-rod	buy	2	The horn's tie-down: an M6 A2 threaded rod about 110 long, an M6 knurled thumb nut and a washer, and a flanged M6 threaded insert (body 9 x 10) for the top's underside: one for plastics in HDPE, bonded with epoxy in acrylic		8.00
36	land-left, land-right	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	25 x 840	9.98
37	land-top, land-bottom	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	470 x 25	5.58
38	back-gasket	buy	2	EPDM sponge strip 25 x 3, closed cell, cut and joined as a frame on the back's four land strips, punched for its 16 screws		8.00
39	back-screws	buy	2	16 M5 button heads (ISO 7380, A2 stainless, black: round 16) on black nylon washers: the back to the inserts in its land strips (M5 x 40 into HDPE or acrylic, M5 x 30 into plywood)		8.00
40	feet	buy	2	Levelling feet M8, a 40 mm swivel base on a rubber pad, black, with lock nuts, a set of 4		18.00
41	port-screws	buy	2	The ports' fixings: eight black countersunk screws, four through each port's frame into the panel (thread-forming screws for plastics, EJOT PT K40 kind, 4 x 16 in the 20 mm HDPE; M4 x 20 into tapped threads in the 25 mm acrylic; wood screws 4 x		5.00
42	top-pad	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 x 8 ft sheet	108 x 28	1.44
43	lining-bottom-1	cut	2	100 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	201 x 295	1.48
44	lining-left-2a, lining-right-2a	cut	4	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	167 x 479	1.00
45	lining-left-2b, lining-right-2b	cut	4	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	144 x 479	0.86
46	lining-left-3, lining-right-3	cut	4	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	332 x 139	0.58
47	lining-top-3	cut	2	100 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	418 x 295	3.08
48	lining-back-a, lining-back-d	cut	4	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	68 x 789	0.68

Parts list (2 of 3)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
49	lining-back-b	cut	2	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	239 x 278	0.83
50	lining-back-c	cut	2	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	239 x 118	0.35
51	woofer-fixings	buy	2	M6 brass threaded inserts, knurled, bonded with epoxy into 8.5 holes in the acrylic panel, with M6 x 25 button heads (A2), a set of 8 (the 15PR400's eight holes on 375)		14.00
52	hf-fixings	buy	2	The 2 in driver's fixings: four M6 x 16 countersunk screws (A2), the driver to its adaptor ring, and four M5 x 16 socket caps with washers, the ring to the brass inserts in the horn's boss, a set		6.00
53	cement	buy	2	Two-part polymerising acrylic cement for thick cast sheet (Acrifix 1R 0192 kind), clear, about 0.5 kg a speaker: the corners, the braces, the ribs and the back's land strips		45.00
54	gasket	buy	4	1.5 mm closed-cell foam gasket strip, 1 m		5.00
55	lining-sleeves	buy	2	White polyester voile (a sheer curtain fabric), about 3 m2 a speaker with its seams, sewn into a sleeve round each batt of the lining, so no fibre reaches the ports or the woofer and the batt's face stays even		10.00
56	lining-tape	buy	2	Clear acrylic foam tape (3M VHB 4910 or alike), 20 mm dots, about 60 a speaker: on a Cloud cabinet four hold each batt's sleeve flat on the acrylic, all but invisible through it		12.00
57	amp-fixings	buy	2	Hypex 4.3 x 16 self-tapping pan heads and 3 mm EPDM tape, a plate's set		10.00
58	amp-box-fixings	buy	2	Ten screws for the amplifier compartment's flange, with washers: thread-forming screws for plastics (EJOT PT K40 x 12 kind) into HDPE, M4 screws into inserts bonded in acrylic, or wood screws into plywood		3.00
59	hf-capacitor	buy	2	33 uF 100 V polypropylene film capacitor, in series ahead of the L-pad (a first-order high-pass at about 300 Hz into 16 ohm)		12.00
60	loom	buy	2	The loom: screw-mount cable-tie bases with thread-forming screws for plastics (EJOT PT K30 x 10 kind) every 120, ties, a polarised 2-way connector (Molex Mini-Fit Jr or Wago 2092 kind) for each lead at the compartment, crimped insulated 6.3		18.00
61	terminals	buy	8	6.3 mm insulated push-on terminal		0.50
62	back-inserts	buy	2	M5 brass threaded inserts, knurled, a set of 16, bonded with epoxy into 8 holes 12 deep in the back's acrylic land strips		10.00
63	feet-inserts	buy	2	M8 brass threaded inserts, knurled, a set of 4, bonded with epoxy into 12.5 holes 15 deep in the acrylic bottom, for the feet		5.00

Parts list (3 of 3)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-10-15 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. The waveguide stands on four posts, stainless lamp tube 10 x 7, each threaded M10 x 1 at its top into a brass insert bonded in a pad cast under the horn (drilled after casting, through a jig taken off the top's sockets), its plain foot standing in a polyurethane sleeve in a brass socket bonded with epoxy into the top, so the horn and its posts lift off as one. An M6 rod down the free rear post, from a thumb nut inside the pod to an insert under the top, holds them down. The cap closing the pod is held by a twist-lock (round 16): three lugs on its spigot in L-slots in the pod's bore, locked by an eighth of a turn, an O-ring beyond them; no screw shows. The HF lead, a round silicone cable 5 across, runs from its connector at the amplifier's compartment straight up, through a gland under the top and up inside a rear post into the pod, where it plugs into the driver's leads. To lift the horn off: the cap off, the nut undone and the lead unplugged.
2. The FA122 crosses at about 800 Hz (LR4: B&C asks 12 dB/oct or steeper there) with the HF channel about 9 dB down (108 dB at 2.83 V against the woofer's 99), part of it in the passive L-pad after the amplifier, so its hiss stays inaudible: two 25 W aluminium-housed resistors on a plate in the compartment, behind a 33 uF film capacitor that keeps low frequencies off the driver if the preset is ever lost.
3. The amplifier's preset, stored before the speaker ships: the crossover; a 4th-order Butterworth high-pass at 28 Hz on the woofer (below the ports' tuning the box no longer holds the cone, which would pass Xmax below about 30 Hz at full output, and a record's warp drives it there); limiters set to the woofer's Xmax and the driver's rating; a notch at the ports' pipe resonance (about 540 Hz in the pressed cabinet, 560 in the Cloud's, 600 in the plywood one), measured in each port's near field.
4. The amplifier in a sealed compartment of its own, as Hypex asks: 1.5 mm aluminium, black anodised, on an EPDM gasket on the back's inner face, so the box stays airtight and the mains electronics are enclosed in metal. The leads leave it through cable glands. A rating plate beside the amplifier carries the model, the serial number, the ratings, the marks and the maker's address; the finished speaker needs an EMC pre-scan and an IEC 62368-1 assessment.
5. The lining: polyester batts (thermally bonded, recycled PET), 50 mm on the sides and 100 mm on the top and the bottom (round 21), one to each bay between the braces (and the ribs), and 50 mm on the back, its full height between its ribs, so they come off with it; each cut 10 full each way and sewn into a sleeve of white voile; none on the front panel, none beside the ports (10 clear of each tube, nothing between its inner end and the back's batts) and none within 20 of the amplifier's compartment.
6. The back comes off (round 14): it closes on a 3 mm EPDM gasket against four land strips inside the shell, sixteen M5 button heads through it into inserts in them. The amplifier's compartment, the back's ribs and its batts come off with it and the leads unplug at the compartment, so the amplifier, the L-pad and the leads can be reached.
7. Four levelling feet (M8, 40 swivel bases on rubber pads) in inserts in the bottom, 50 in from its edges, lift the cabinet 12 mm: set them so it stands level and cannot rock under the horn, and lock their nuts.
8. No wood: every panel is cast acrylic, 20 mm, tinted through in its colour (red, orange, amber, green, blue, violet, frosted, black), recycled (made from recycled MMA) where a maker can make the colour, cut on a CNC router from 4 x 8 ft sheets of one batch (approve a tint swatch and a cemented corner sample first). The ring (6 mm) is cast acrylic tinted opaque in the ring colour; the cloth disc's frame (12 mm) black cast acrylic.
9. Assembly: each panel's edges milled square and polished where they show, the panels annealed (about 80 C, an hour for each 3 mm of thickness, cooled slowly); the ribs cemented to their panels flat (the back's to the back); the shell's corners, the front panel (set back 24 mm), the two braces and the back's four land strips cemented with a two-part polymerising acrylic cement for thick sheet (Acrifix 1R 0192 or alike), clamped square on a jig; the box annealed again; the back closes the box last, on its gasket and sixteen screws, and comes off. No solvent cement: it crazes thick cast sheet.
10. The woofer flush in its rebate on a gasket, on eight M6 brass inserts bonded with epoxy in 8.5 holes; the ring over its frame on its magnets, bonded in their pockets; each port in the cabinet's own acrylic, its tube cemented into its 8 mm frame, fitted from the front into a recess on a foam gasket, four black M4 screws in its corners into tapped threads; the tubes cut to the length found on the first sample before their ends are flared.
11. The front: the panel's own tinted acrylic, the lining showing through it round the woofer, and a cloth disc over the woofer in the acrylic's colour, its frame standing on the ring on four magnet pairs.

Notes

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

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Notes

12. Seen through: everything inside is drawn to be seen. The leads run straight and tied, the glands and the brass sockets show, the amplifier's compartment is black anodised, and each batt's voile sleeve has its seams to the wall.

Notes

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